

Forget to remember
Remember to forget

Long Short Term Memories and Gated Recurrent Units

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Vision, Learning and Control
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Some of the images and animations used here were originally designed by Adam Prügel-Bennett.

Recap: An RNN is just a recursive function invocation

- $\mathbf{y}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{c}(t) | \mathbf{W}_f)$
- and the state $\mathbf{c}(t) = \mathbf{g}(\mathbf{x}(t), \mathbf{c}(t-1) | \mathbf{W}_g)$
- If the output $\mathbf{y}(t)$ depends on the input $\mathbf{x}(t-2)$, then prediction will be

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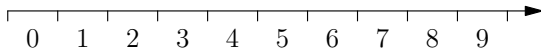
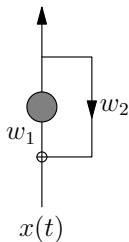
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- This either vanishes or explodes when τ becomes large

Vanishing and Exploding Gradients

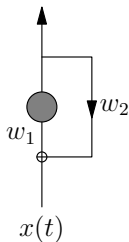
$$y(t) = w_1 (x(t) + w_2 y(t-1))$$



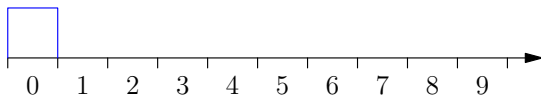
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$$y(t) = w_1 (x(t) + w_2 y(t-1))$$

$$w_1 = w_2 = 0.9$$



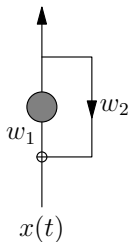
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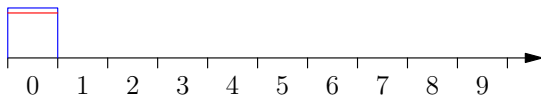
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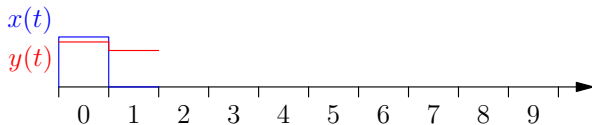
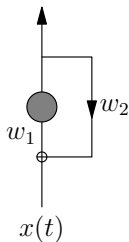
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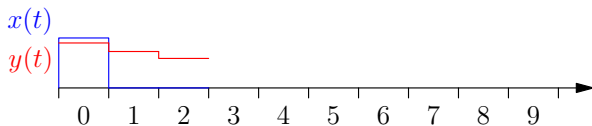
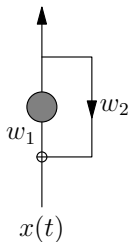
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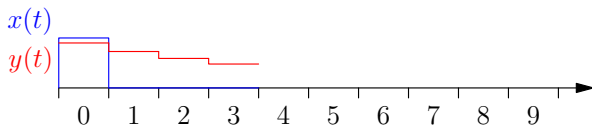
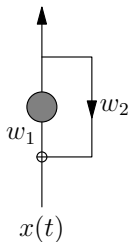
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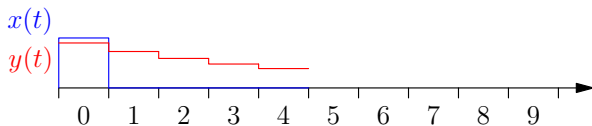
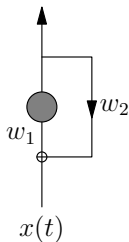
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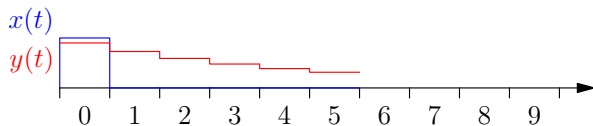
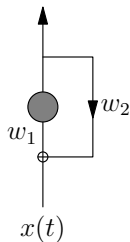
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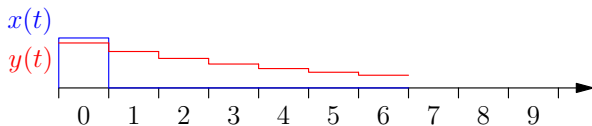
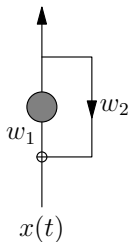
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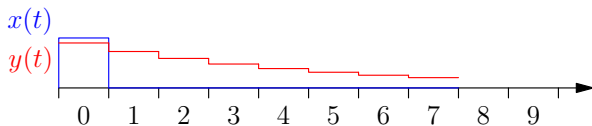
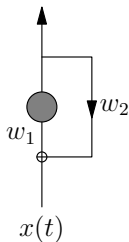
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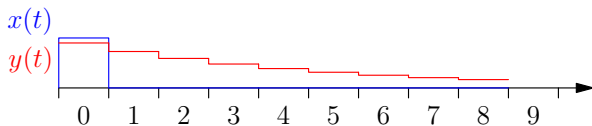
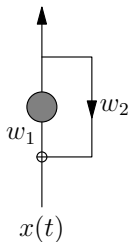
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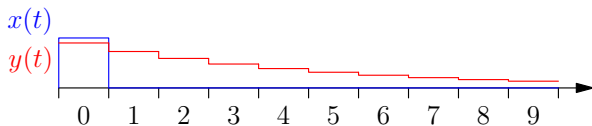
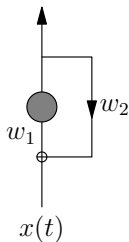
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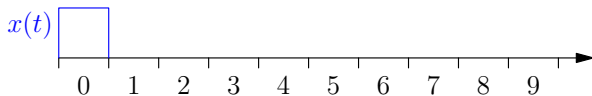
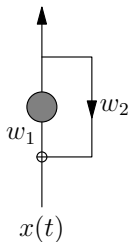
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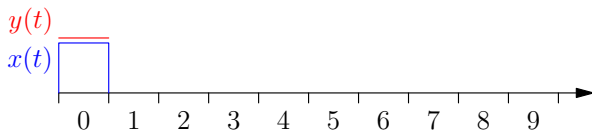
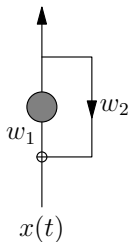
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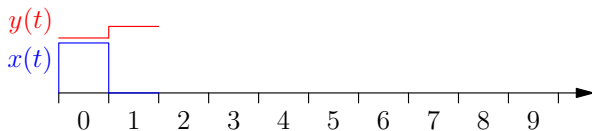
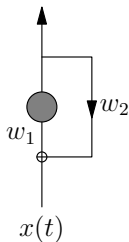
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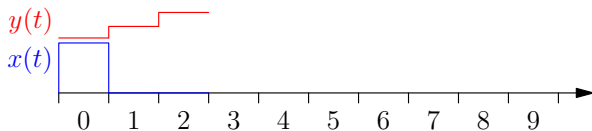
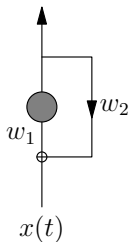
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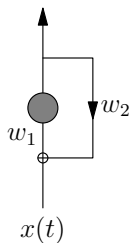
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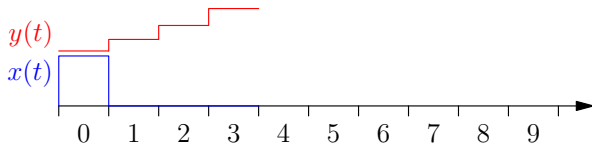


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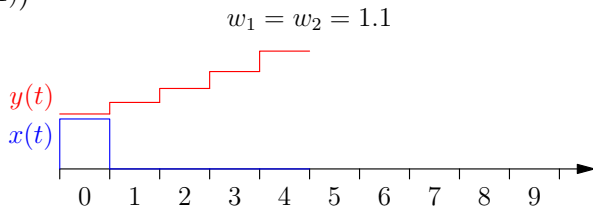
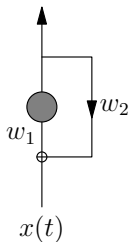


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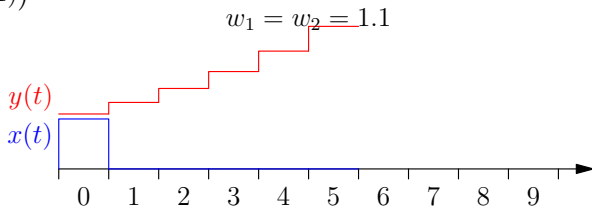
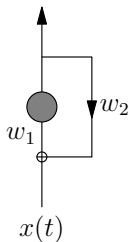
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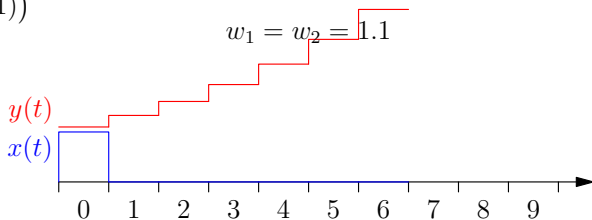
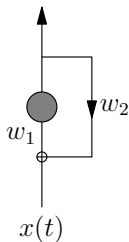
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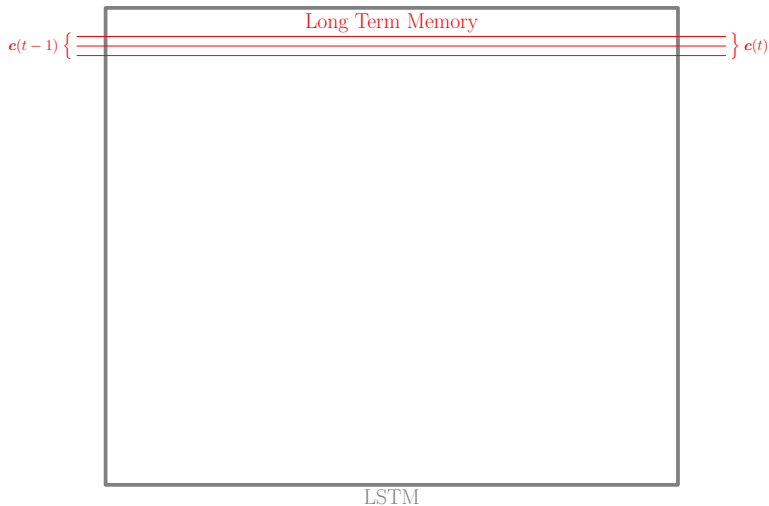
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- Sigmoid functions naturally saturate at 0 and 1

LSTM Architecture

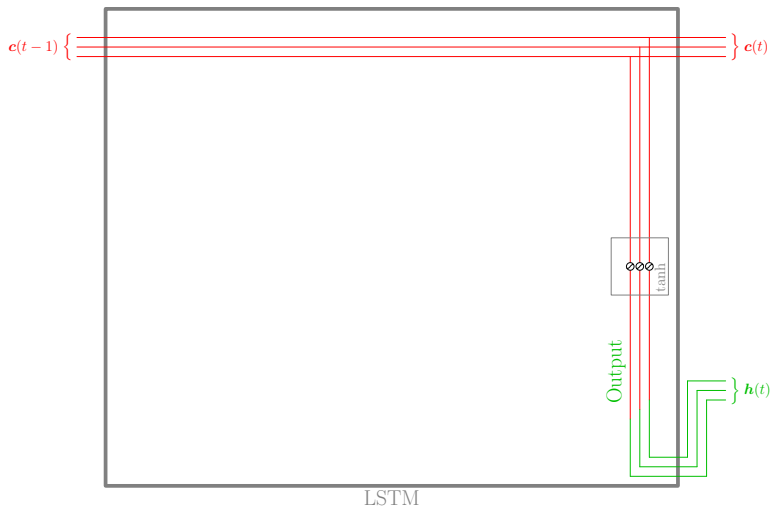


LSTM

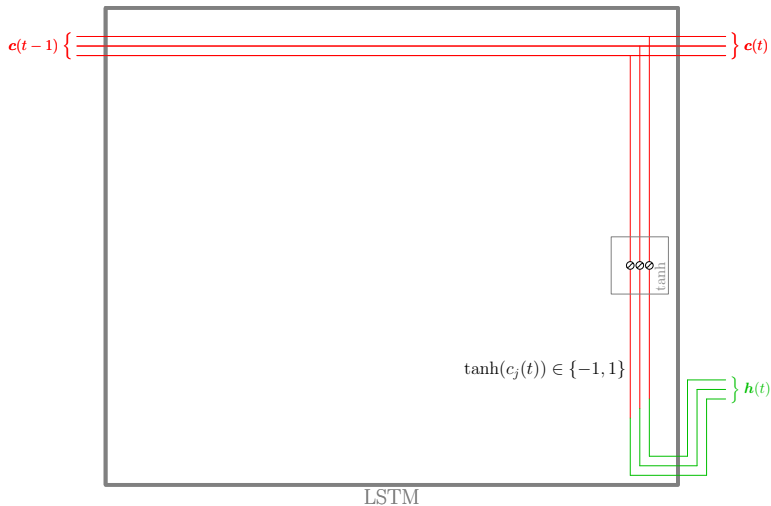
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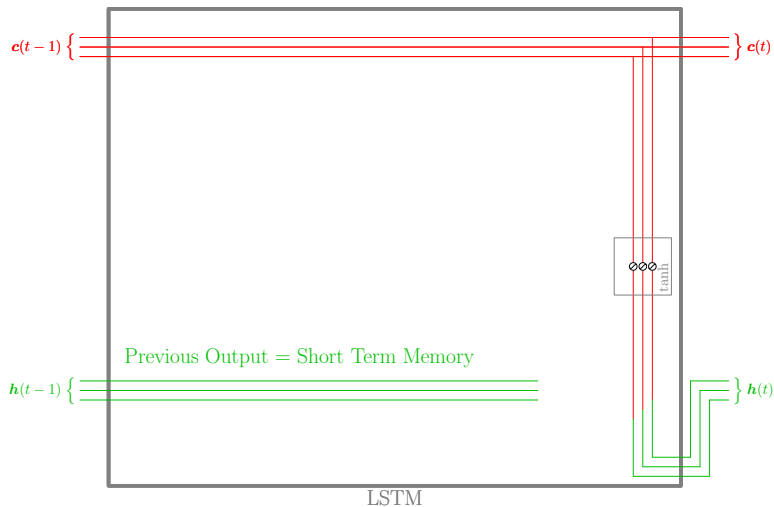
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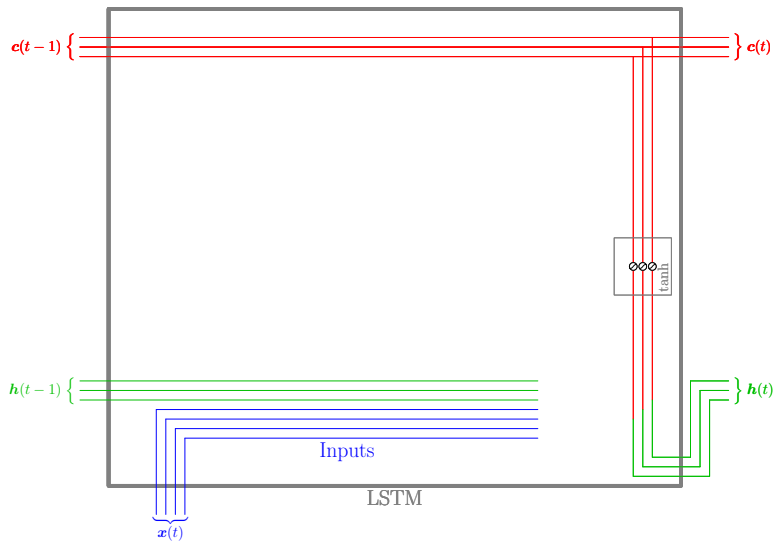
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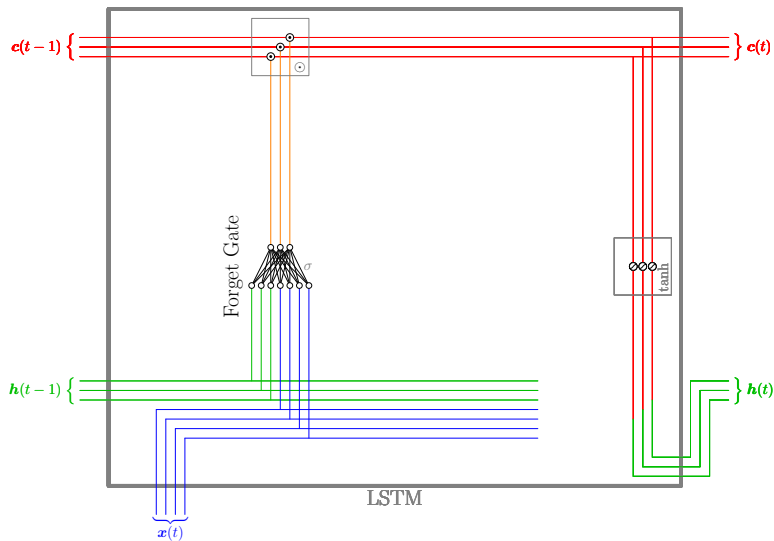
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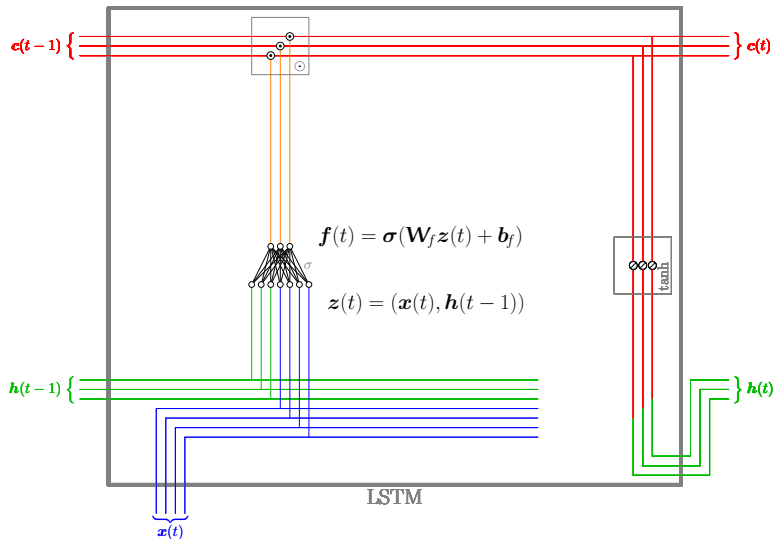
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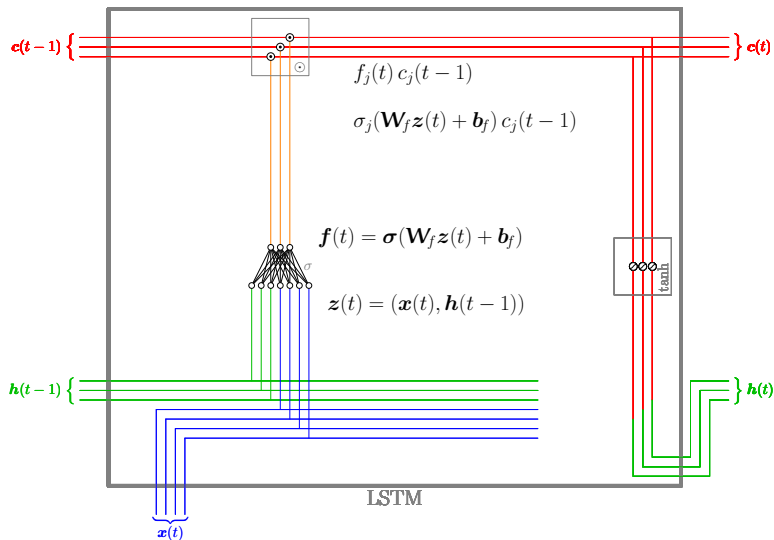
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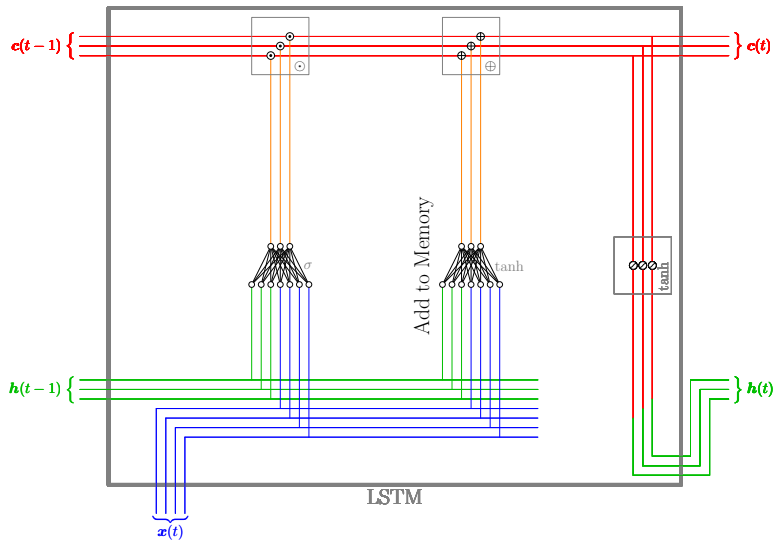
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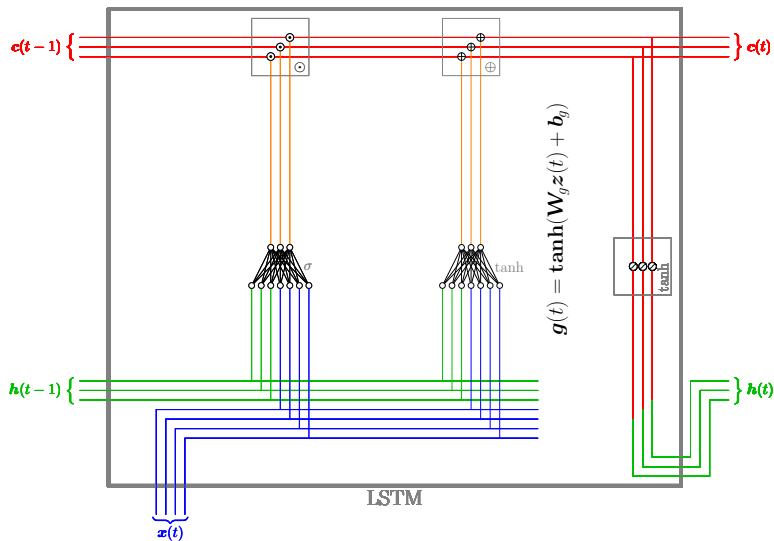
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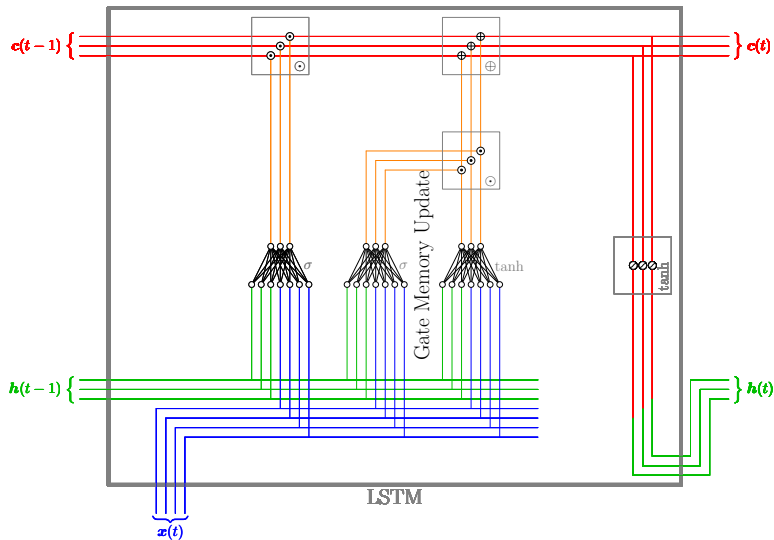
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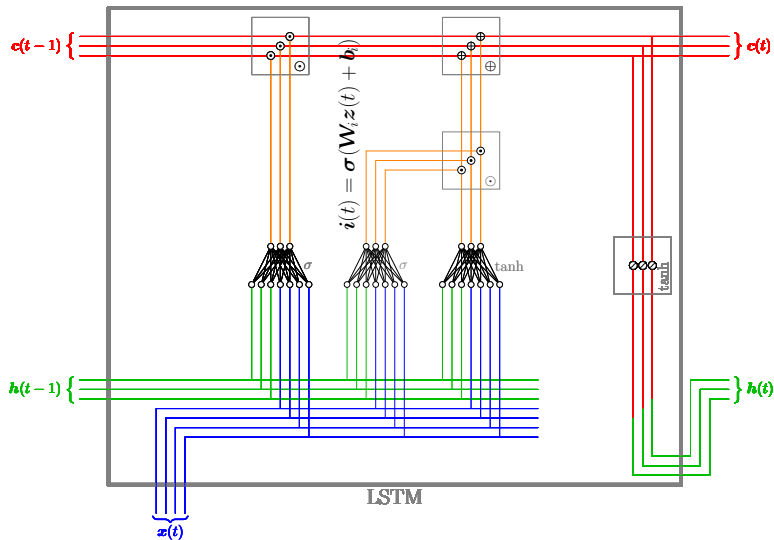
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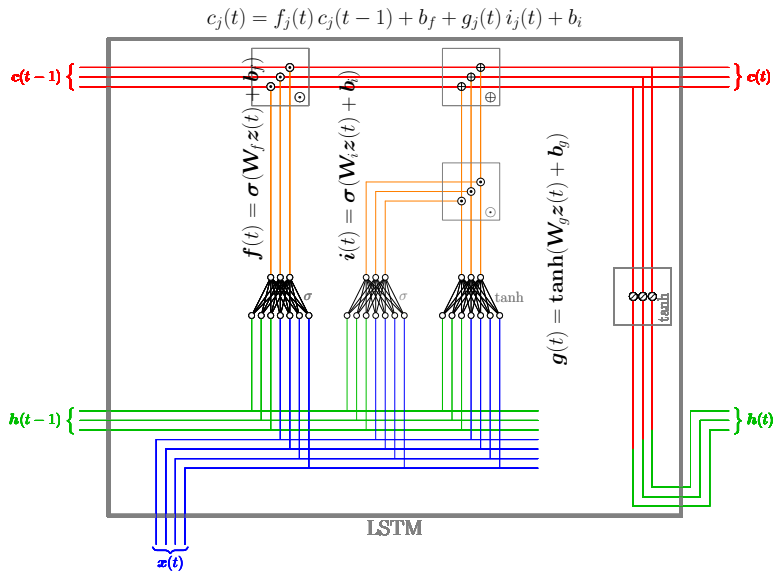
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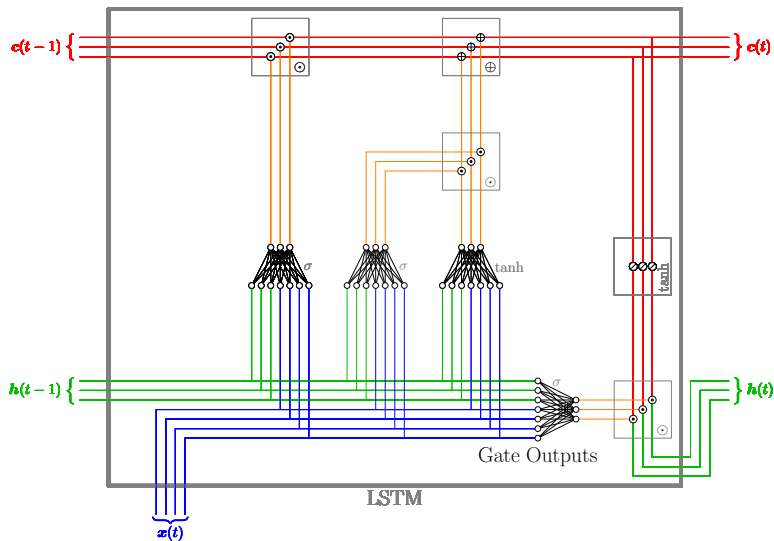
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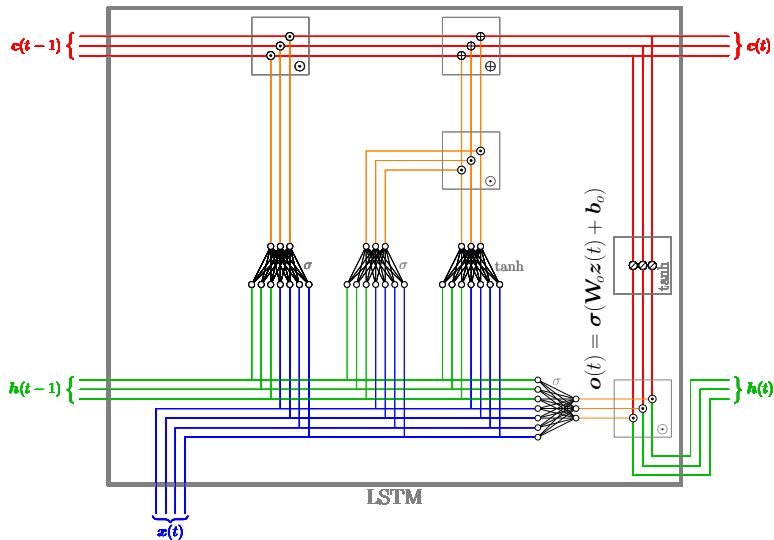
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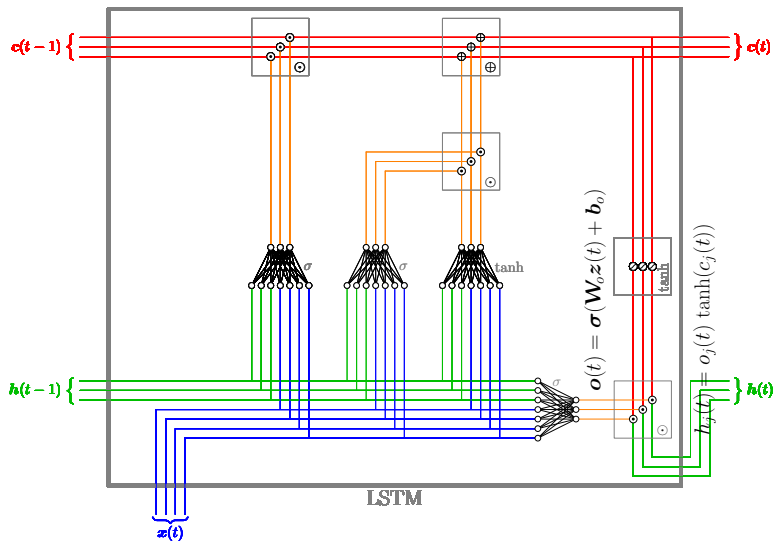
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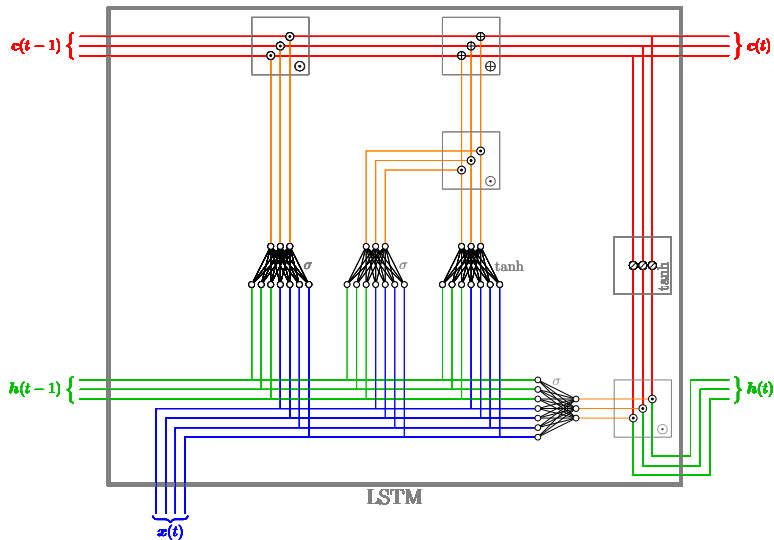
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Update Equations

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- Long-term memory update

$$\mathbf{c}(t) = \mathbf{f}(t) \odot \mathbf{c}(t-1) + \mathbf{g}(t) \odot \mathbf{i}(t)$$

Update Equations

Initially, for $t = 1$, $\mathbf{h}(0) = \mathbf{0}$

- Inputs $\mathbf{z}(t) = (\mathbf{x}(t), \mathbf{h}(t-1))$
- Network updates ($\mathbf{W}_*$ and $\mathbf{b}_*$ are the learnable parameters)

$$\begin{aligned}\mathbf{f}(t) &= \sigma(\mathbf{W}_f \mathbf{z}(t) + \mathbf{b}_f) & \mathbf{i}(t) &= \sigma(\mathbf{W}_i \mathbf{z}(t) + \mathbf{b}_i) \\ \mathbf{g}(t) &= \tanh(\mathbf{W}_g \mathbf{z}(t) + \mathbf{b}_g) & \mathbf{o}(t) &= \sigma(\mathbf{W}_o \mathbf{z}(t) + \mathbf{b}_o)\end{aligned}$$

- Long-term memory update

$$\mathbf{c}(t) = \mathbf{f}(t) \odot \mathbf{c}(t-1) + \mathbf{g}(t) \odot \mathbf{i}(t)$$

- Output $\mathbf{h}(t) = \mathbf{o}(t) \odot \tanh(\mathbf{c}(t))$

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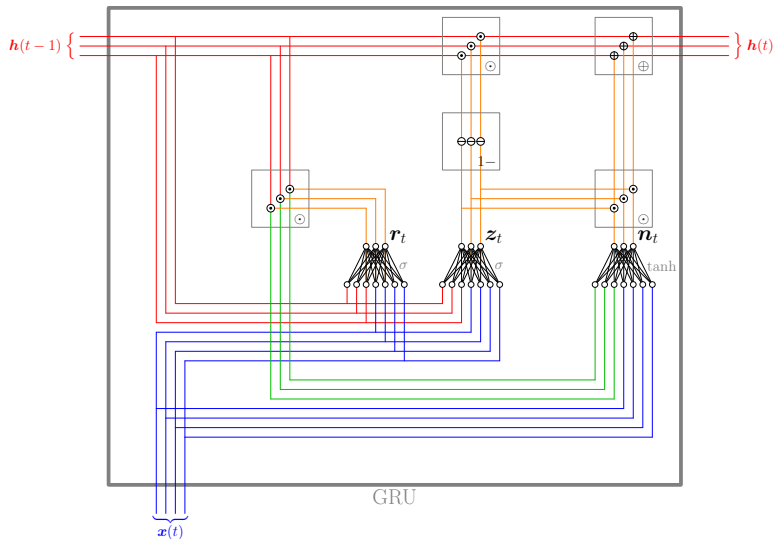
- We can train an LSTM by unwrapping it in time.
- Note that it involves four dense layers with sigmoidal (or tanh) outputs.
- This means that typically it is very slow to train.
- There are a few variants of LSTMs, but all are very similar. The most popular is probably the Gated Recurrent Unit (GRU).

LSTM Success Stories

- LSTMs have been used to win many competitions in speech and handwriting recognition.
- Major technology companies including Google, Apple, and Microsoft are using LSTMs as fundamental components in products.
- Google used LSTM for speech recognition on the smartphone, for Google Translate.
- Apple uses LSTM for the "Quicktype" function on the iPhone and for Siri.
- Amazon uses LSTM for Amazon Alexa.
- In 2017, Facebook performed some 4.5 billion automatic translations every day using long short-term memory networks¹.

¹https://en.wikipedia.org/wiki/Long_short-term_memory

Gated Recurrent Unit (GRU)



Gated Recurrent Unit (GRU)

- $\mathbf{x}(t)$: input vector
- $\mathbf{h}(t)$: output vector (and 'hidden state')
- $\mathbf{r}(t)$: reset gate vector
- $\mathbf{z}(t)$: update gate vector
- $\mathbf{n}(t)$: new state vector (before update is applied)
- $\mathbf{W}$ and $\mathbf{b}$: parameter matrices and biases

Gated Recurrent Unit (GRU)

Initially, for $t = 1$, $\mathbf{h}(0) = \mathbf{0}$

$$\mathbf{z}(t) = \sigma(\mathbf{W}_z(\mathbf{x}(t), \mathbf{h}(t-1)) + \mathbf{b}_z)$$

$$\mathbf{r}(t) = \sigma(\mathbf{W}_r(\mathbf{x}(t), \mathbf{h}(t-1)) + \mathbf{b}_r)$$

$$\mathbf{n}(t) = \tanh(\mathbf{W}_n(\mathbf{x}(t), \mathbf{r}(t) \odot \mathbf{h}(t-1)) + \mathbf{b}_h)$$

$$\mathbf{h}(t) = (1 - \mathbf{z}(t)) \odot \mathbf{h}(t-1) + \mathbf{z}(t) \odot \mathbf{n}(t)$$

Most implementations follow the original paper and swap $(1 - \mathbf{z}(t))$ and $(\mathbf{z}(t))$ in the $\mathbf{h}(t)$ update; this doesn't change the operation of the network, but does change the interpretation of the update gate, as the gate would have to produce a 0 when an update was to occur, and a 1 when no update is to happen (which is somewhat counter-intuitive)!

GRU or LSTM?

- GRUs have two gates (reset and update) whereas LSTM has three gates (input/output/forget)
- GRU performance on par with LSTM but computationally more efficient (less operations & weights).
- In general, if you have a very large dataset then LSTMs will likely perform slightly better.
- GRUs are a good choice for smaller datasets.